

The Saturation Effect Analysis of Oriented Silicon Steel Permanent Magnet Linear Synchronous Motor

Ting Dong*, Xuepeng Wei, Yansheng Bai, Yuan Zhang

School of Electrical Engineering, Shenyang University of Technology, Shenyang 110870, Liaoning, China

Email: t_dong_sut@126.com

Abstract — Due to the nonlinear influence of ferromagnetic materials, the saturation effect of the magnetic circuit becomes one of the main factors affecting the improvement of motor performance. Magnetic saturation is one of the major challenges for all electrical motors and makes it much more difficult to design high performance electrical motors. The magnetic permeability of the oriented silicon steel, which is often used in transformers, is significantly higher than that of the traditional silicon steel, and can provide a higher magnetic flux density for the motor. Firstly, the influence of the saturation effect of the magnetic circuit on the inductance parameters is analyzed. Secondly, using the finite element software Ansoft, the inductance parameters of the traditional silicon steel and oriented silicon steel PMLSMs are calculated with different loads. Simulation results show that the inductance of the surface-mounted PMLSM is no longer a constant due to the influence of the armature reaction with load. The magnetic saturation of the oriented silicon steel PMLSM decreases by 4.4% and 19% respectively at rated load and 5 times rated load, and its overload capacity is stronger.

Key Words: Oriented silicon steel, PMLSM, Inductance

I. INTRODUCTION

Compared with other linear motors, permanent magnet linear synchronous motors (PMLSM) have the advantages of large thrust, low loss, high response speed, and high precision. They have been widely used in high-precision computer numerical control (CNC) motor tools, vertical lift systems, wave energy generators, and transportation systems[1-3]. There are many pole forms with PMLSM, and the surface-mounted type is discussed in this paper, as shown in Fig.1.



Fig. 1 Surface-mounted PMLSM

Phase inductance is one of the most important parameters of PMLSM. It not only reflects the degree of motor saturation directly, but also affects the control accuracy of the motor. At present, the prediction of permanent magnet motor inductance is generally divided into analytical method and finite element method. In Ref. [4], an equivalent magnetic circuit was used to predict the inductance. First, the magnetic circuit of the motor was segmented, and then the magnetic resistance of each segment was considered

separately. However, it is assumed that the stator yoke is unsaturated, which is inconsistent with the actual situation. It will inevitably produce errors. Since the analysis method cannot generally consider the influence of magnetic circuit saturation and leakage flux. And the traditional analytical method generally considers the inductance to be a constant, which is also inconsistent to the finite element calculation result, so there is a large error. In Ref. [5-7], a frozen permeability (FP) method based on finite element is proposed to separate the permanent magnet field and the armature magnetic field linearly. In Ref. [6], the d-axis and q-axis inductances are predicted based on FP and stored magnetic energy method to improve accuracy of torque estimation in a 36-slot/4-pole (36s/4p) interior permanent magnet motor. In Ref. [8-9], the variation of the permanent magnetic flux and inductance with the load was calculated by FP, and the inductance is the function of the rotor position. In Ref. [10], the d-axis and q-axis inductance of the synchronous reluctance motor are calculated by the equivalent circuit model, and the iron loss and saturation are considered. In Ref. [11], a generalized dynamic model of self-excited induction generator is presented. The model accounts for the effects of magnetic saturation by expressing the magnetizing inductance as an exponential function of the magnetizing current. In Ref. [12], the inductance value is used to measure the saturation effect of the reactor by finite element method. Inductance is the main parameter for analyzing the saturation of magnetic circuit. Although there are various methods for predicting the inductance parameters, predict the saturation effect, few literatures apply the inductance parameters to analyze the saturation effect of magnetic circuit.

In this paper, the oriented silicon steel is applied to the mover core of PMLSM, and the saturation effects of PMLSM with different materials are analyzed by phase inductance respectively. The finite element software Ansoft is used to calculate the inductance parameters of the traditional PMLSM and the proposed PMLSM, and analyzes the variation of the inductance parameters with the position of the mover. The results verify that the oriented silicon steel has a significant effect on improving the motor saturation.

II. ORIENTED SILICON STEEL PMLSM

The PMLSM, used in the field of direct-drive, are usually designed with similar number of poles and slots structure. The previous research shows that the magnetic circuit characteristics is different with the traditional integral-slot PMLSM. It has a longer magnetic circuit of the teeth part and a shorter magnetic circuit of the yoke part, and these characteristics provide the conditions for the application of the oriented silicon steel. The oriented silicon steel is an anisotropic ferromagnetic material. Although its

permeability in the cutting direction is low, the permeability in the rolling direction is higher than that in the traditional silicon steel, and it is generally applied to a transformer core. The BH curves of the traditional silicon steel DW465 and rolling direction of the oriented silicon steel B30P120 are shown in Fig.2. It can be see obviously that the knee point and saturation point of the oriented silicon steel are both higher than the points of the traditional silicon steel (non-oriented).

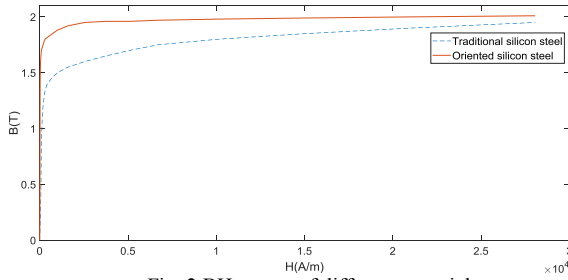


Fig. 2 BH curves of different materials

Therefore, as shown in Fig.3, making the rolling direction of the oriented silicon steel coincides with the direction of the tooth magnetic circuit, so the longer teeth magnet circuits utilize the high permeability in the rolling direction of the oriented silicon steel. On the other hand, the yoke part is short, the yoke is usually not easy to saturate in the motor with similar number of poles and slots, so the lower permeability in the cutting direction will not cause bad effect. So the proposed structure can improve magnetic circuit saturation.

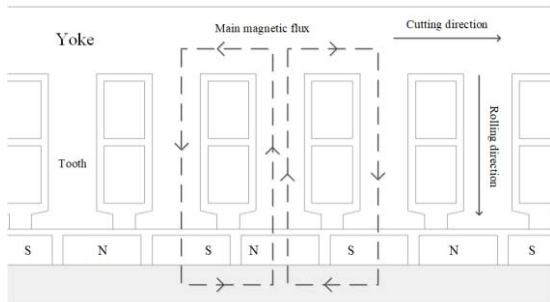


Fig.3 The main magnetic flux of similar number of poles and slots PMLSM

III. THE SATURATION EFFECT ANALYSIS OF ORIENTED SILICON STEEL

Taken a 12-pole/14-slot PMLSM as example, the finite element models of traditional silicon steel PMLSM and oriented silicon steel PMLSM have been constructed, as shown in Fig.4. The two models have the same dimensions but different mover materials. The main parameters are shown in the Tab.1. The moving direction of the oriented silicon steel PMLSM is the cutting direction of the oriented silicon steel sheet and the direction of the tooth magnetic circuit is the rolling direction of the oriented silicon steel sheet.

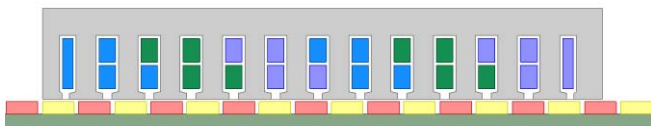


Fig.4 Simulation model

TABLE 1

Main Parameters of the Motor	
Parameter(mm)	Value
Pole Pitch	12
PM Width	10.5
PM Height	4
Tooth Width	6.5
Tooth Height	21
Slot Width	7.5
Motor Width	77

A. The Saturation Effect Analysis of Oriented Silicon Steel

According to the characteristics of the AC magnetic circuit, the inductance of the core coil is expressed as:

$$L = N^2 \Lambda_m = \frac{N^2}{R_m} \quad (1)$$

Where N , Λ_m and R_m are the number of turns, magnetic permeance and magnetic reluctance respectively.

With the different loads, the saturation effects of the magnetic circuit are significantly different. When the load increases, the armature current must to increase to satisfy the load requirement, as a result of that, the magnetic field intensity increases. However, due to the non-linear characteristics of ferromagnetic materials, as the magnetic field intensity increases, the ferromagnetic material tends to saturate, the magnetic permeability decreases, and R_m increases, inevitably causing the inductance parameters to decrease. The more significant the saturation effect of the magnetic circuit, the smaller the inductance parameter, so this paper uses the inductance parameter to discuss the saturation of the magnetic circuit.

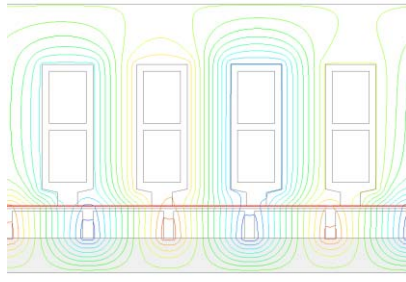
B. Finite Element Calculation of Inductance Parameters

Finite element calculation model can set the nonlinear and saturation factors of ferromagnetic materials by inputting B-H curves, and obtain calculate the electromagnetic field distribution accurately, so it is widely used. Sometimes the 3D finite element model is used to calculate the inductance parameters, in order to take the end inductance of the winding into consideration, but the 3D finite element model needs a large amount of calculation and takes more time. Usually the end inductance only accounts for about 10% of the total phase inductance. Therefore, 2D finite element model is adopted to discuss the saturation problem.

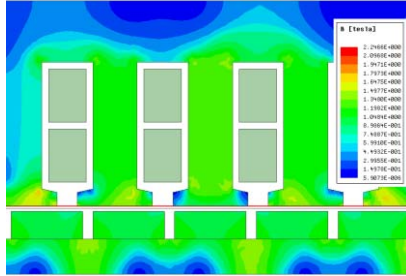
C. No-load analysis of Non-oriented silicon steel PMLSM

Without load, the armature current is equal to 0, and the magnetic field is entirely generated by the permanent magnet. The no-load characteristics of the non-oriented silicon steel sheet PMLSM mainly include the no-load magnetic line distribution and the no-load magnetic density distribution. By analyzing the no-load characteristics of the motor, it is possible to analyze the saturation of the magnetic field and whether the magnetic circuit design is reasonable.

The no-load magnetic field distribution and no-load magnetic density distribution of the non-oriented silicon steel sheet PMLSM are shown in Fig.5 (a) and (b) respectively.



(a) No-load magnetic force lines



(b) No-load magnetic densities

Fig.5 No-load magnetic field with non-oriented silicon steel PMLSM

It can be seen from the distribution of the no-load magnetic force lines that the main flux distribution conforms to the magnetic circuit characteristics and the leakage flux is less, and the magnetic field distribution is reasonable and not saturated.

When the non-oriented silicon steel sheet PMLSM without load, the A phase inductance is shown in Fig.6. Without load, the saturation of the motor is low, and because of surface-mounted structure, the A phase inductance changes very little and can be regarded as a constant.

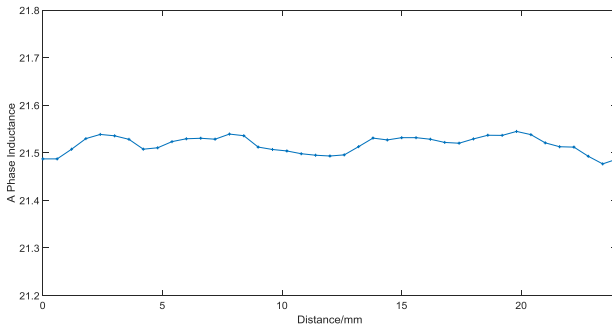
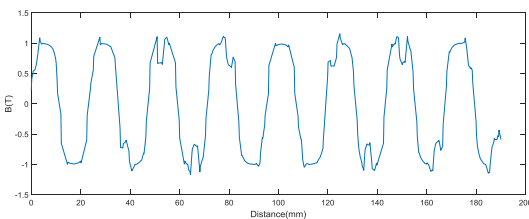


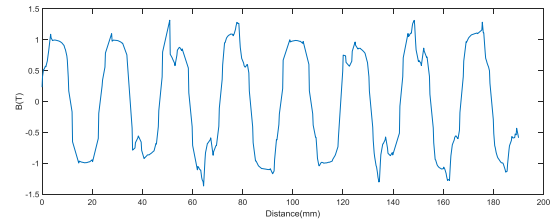
Fig.6 No-load A phase inductance

D. Load analysis of Non-oriented silicon steel PMLSM

Through calculating the no-load magnetic field and rated load magnetic field, the air-gap magnetic density distributions are obtained, as shown in Fig.7 (a) and (b) respectively. Under the influence of the armature reaction, the air gap flux density waveforms are significant distorted obviously.

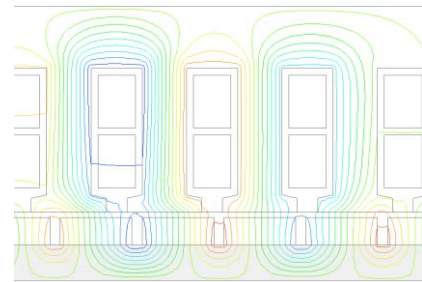


(a) Air gap magnetic density without load

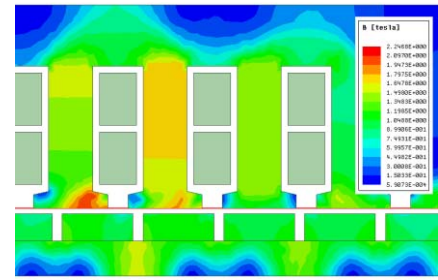


(b) Air gap magnetic density with rated load
Fig.7 Air gap magnetic density of the Prototype

At rated load, the magnetic force lines and magnetic densities are shown in Fig.8 (a) and (b) respectively. Due to the influence of the armature reaction, the magnetic force lines in the load magnetic field are no longer symmetrical and local saturation phenomenon occurs.



(a) Load magnetic force lines



(b) Load magnetic densities
Fig.8 Prototype magnetic field

E. Analysis of Inductance Parameters of Non-oriented Silicon Steel PMLSM

The model adopts three phase symmetrical winding, so one phase winding inductance can reflect the saturation situation of magnetic circuit. Here taken A-phase winding for example, in order to obtain the relationship between the phase inductance parameters and the position of the mover, one pair pole pitch (24mm) is selected for discretization. Setting the A phase winding axis coincides with the d-axes in the initial position, A-phase winding inductance is shown in Fig.9.

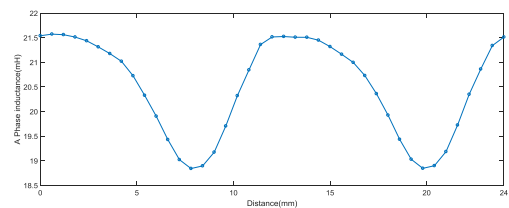


Fig.9 Rated load A-phase inductance

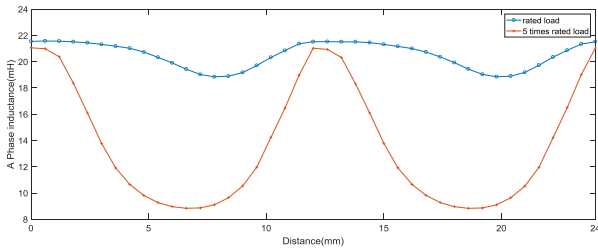
When the motor is running at rated load, the A phase inductance forms two peaks and two troughs within the distance of a pair of magnetic poles. In the peak position, the magnetic potential of the A-phase winding is opposite to the

magnetic potential of the permanent magnet, and demagnetization is performed. Therefore, the saturation of the core is the lowest, and the inductance of the A-phase winding is the largest. On the contrary, in the trough position, the saturation of the core is the greatest, and the inductance of the A-phase winding is the smallest.

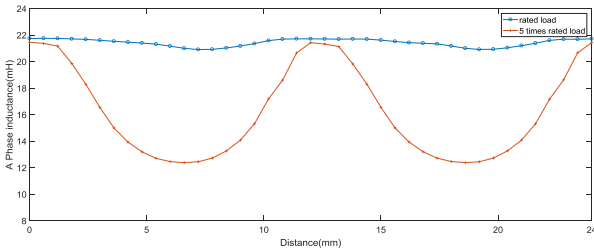
Due to the action of the armature reaction, the motor magnetic circuit is partially saturated. When the armature magnetic field and the permanent magnet magnetic field have the same direction, the magnetic circuit has the highest degree of saturation and the inductance is the smallest.

F. Analysis of Saturation Effect of Non-oriented Silicon Steel and Oriented Silicon Steel PMLSM

Replacing the traditional silicon steel with the oriented silicon steel and setting the different B-H waves in the rolling direction and cutting direction respectively, the A-phase winding inductances of the non-oriented silicon steel and the oriented silicon steel PMLSM with the rated load and 5 times the rated load are obtained respectively, as shown in Fig.10 (a) and (b) respectively.



(a) The inductance of traditional silicon steel PMLSM



(b) The inductance of oriented silicon steel PMLSM
Fig.10 Prototype inductance parameter comparison

The average values of the A-phase inductance are 20.5mH and 21.4mH for the non-oriented silicon steel and oriented silicon steel PMLSM respectively. The average values of A-phase inductances are 13.7mH and 16.3mH respectively at 5 times rated load. Regardless of the traditional silicon steel PMLSM or the oriented silicon steel PMLSM, when the 5 times load is applied, the inductance of the A-phase winding is significantly reduced. From the results, it also proved the feasibility of inductance parameters to measure the saturation of the magnetic circuit.

With rated load, the A-phase winding inductance of the oriented silicon steel PMLSM is reduced by 0.9mH and the magnetic circuit saturation is reduced by 4.4% compared with the non-oriented silicon steel PMLSM. With 5 times the rated load, the A-phase winding inductance of the oriented silicon steel sheet PMLSM is reduced by 2.6mH and the magnetic circuit saturation is reduced by 19%. Therefore, when the oriented silicon steel is applied to the PMLSM, the degree of saturation of the motor can be

significantly reduced, and the advantage of the oriented silicon steel is more obvious when the motor is under overload operation. The overload capability of the oriented silicon steel PMLSM is obviously better than that of the non-oriented silicon steel PMLSM.

IV. CONCLUSION

In this paper, oriented silicon steel sheet is applied to permanent magnet linear synchronous motors to improve the magnetic flux density of the motor and improve the magnetic circuit saturation.

Firstly, the influence of the saturation effect of the magnetic circuit on the inductance parameters is analyzed. The finite element method is used to calculate the electromagnetic field and the armature reaction not only causes the distortion of the air gap flux density waveform, but also causes local saturation phenomenon at rated load. Secondly, the variation of the inductance parameters of non-oriented silicon steel and oriented silicon steel PMLSM under different loads are calculated. The feasibility of measuring the saturation of the magnetic circuit by the inductance parameters is verified. By comparing the inductance of the non-oriented silicon steel and the oriented silicon steel PMLSM at different loads, it is proved that the oriented silicon steel has a significant effect on reducing the saturation of the magnetic circuit, and the oriented silicon steel PMLSM has stronger overload capacity.

V. ACKNOWLEDGEMENT

This work was supported by the National Natural Science Foundation of China under Grant “51607116”.

VI. REFERENCES

- [1] Zhang Z H, Shi L M, Wang K, Li Y H. Characteristics investigation of single-sided ironless PMLSM based on halbach array for medium-speed Maglev train[J]. CES Transactions on Electrical Motors and System, Vol.1, No.4, pp372-382. 2017.
- [2] S G Lee, S A Kim, Zhu Y W, Y H Cho. Optimal Structure Design for Minimizing Detent Force of PMLSM for a Ropeless Elevator[J]. IEEE Transactions on Magnetics, Vol.50, No.1, 2014.
- [3] N. Hodgins, O. Keysan, A. S. McDonald, and M. A. Müller. Design and testing of a linear generator for wave-energy applications[J] IEEE Trans. Ind. Electron., Vol.59, No.5, pp.2094-2103, 2012.
- [4] JB Im, W Kim, K Kim, CS Jin, JH Choi. Inductance Calculation Method of Synchronous Reluctance Motor Including Iron Loss and Cross Magnetic Saturation[J]. IEEE Transactions on Magnetics, 2009, 45(6):2803-2806.
- [5] Hu J, Zou J, Liang W. Finite element calculation of the saturation dq-axes inductance for a direct drive PM synchronous motor considering cross-magnetization[C]. International Conference on Power Electronics and Drive Systems, 2003: 677-681.
- [6] Kang G H, Hong J P, Kim G T, et al. Improved parameter modeling of interior permanent magnet synchronous motor based on finite element analysis[J]. IEEE Transactions on Magnetics, 2000, 36(4): 1867- 1870.
- [7] Williamson S, Knight A M. Performance of skewed single-phase line-start permanent magnet motors[J]. IEEE Transactions on Industry Applications, 1999, 35(3): 577-582.
- [8] Chu W Q. Investigation of torque characteristics of permanent magnet and electrically excited motors[D]. UK: The University of Sheffield, 2013.
- [9] Schmidt E, Susic M. Parameter evaluation of permanent magnet synchronous motors with tooth coil windings using the frozen permeabilities method with the finite element analyses[C]. IEEE Canadian Conference on Electrical & Computer Engineering, Montreal, QC, 2012: 1-5.
- [10] JB Im, W Kim, K Kim, CS Jin, Inductance Calculation Method of Synchronous Reluctance Motor Including Iron Loss and Cross

Magnetic Saturation [J], IEEE Transactions on Magnetics, Vol.5, No.6, pp2803-2806. 2009.

- [11] A Alfarhan, SM Gadoue, B Zahawi, M Shalaby, MA Elgendy, Modelling of magnetizing inductance saturation in self-excited induction generators [J], 2016 IEEE 16th International Conference on Environment and Electrical Engineering (EEEIC), 7-10 June, 2016.
- [12] Filip Grecki, Aleksander Polit, Kacper Sowa, Saturation impact on inductance value of cylindrical reactor [J], 2016 International Symposium on Industrial Electronics (INDEL), 3-5 Nov, 2016.